

2D $U(1)$ Lattice Gauge Theory Simulation & Benchmarks

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1. Overview and Objective

The purpose of this project is to calculate Wilson Lines for a $U(1)$ Lattice Gauge theory and to calculate a physical observable. The observable chosen for this project was

$$\langle s \rangle = \beta(1 - \langle \cos \theta \rangle)$$

(representing the average action per plaquette) due to its simplicity, allowing this to function as a proof of concept.

2. Discretization & Action

- Spacetime has been discretized into an $L \times L = 16 \times 16$ grid
- Plaquettes are defined as

$$U_{\mu\nu}(n) = U_\mu(n)U_\nu(n + \hat{\mu})U_{-\mu}(n + \hat{\mu} + \hat{\nu})U_{-\nu}(n + \hat{\nu})$$

- Gauge links in $U(1)$ are defined as $U_\mu(x) = e^{i\theta_\mu(x)}$ allowing us to write

$$\theta_{\mu\nu}(n) = \theta_n + \theta_{n+\hat{\mu}} - \theta_{n+\hat{\mu}+\hat{\nu}} - \theta_{n+\hat{\nu}}$$

for the entire plaquette angle

- The change in action after a sweep is $\Delta S = -\beta \sum_{\mu,\nu} \cos \theta_{\mu\nu}$

3. Numerical Algorithm & Implementation

- Used Metropolis-Hastings algorithm to decide to accept each step
- Vectorized link candidate generation using NumPy broadcasted operations on $L \times L$ link arrays
- Checked thermalization by monitoring plaquette energy history over 10^3 Monte Carlo sweeps

4. Key Results

- A very clear trend was able to be obtained when comparing the observable $\langle s \rangle$ in terms of our coupling constant β shown in Figure 2
- Although there is a large amount of noise present in Figure 1, it is clear that it plateaus very quickly around $\langle \cos \theta \rangle \approx 0.45$ demonstrating that the algorithm worked successfully

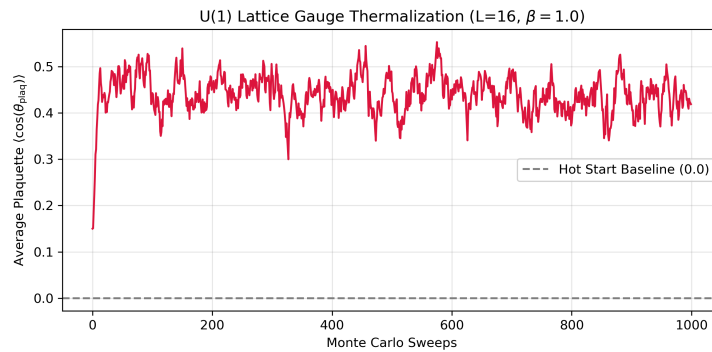


Figure 1: Thermalization Plot in Matplotlib

Average Plaquette Action based on varying coupling constants β for a U(1) Gauge Theory (L=16, $\varepsilon = 0.5$)

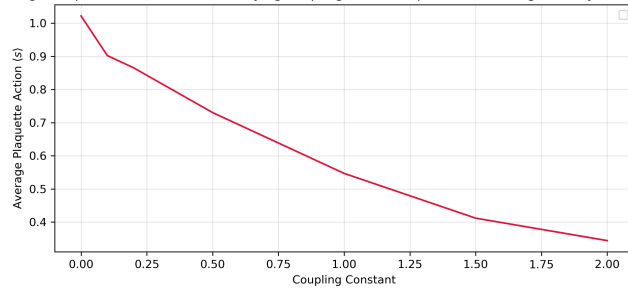


Figure 2: Observable yielding clear trend W.R.T. β